

SATYENDRA

ROBOTICS ENGINEER — MOTION PLANNING, CONTROLS & SYSTEM INTEGRATION

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Portfolio: <https://sahnisatyendra022-cloud.github.io/portfolio-page/#projects>

PROFESSIONAL SUMMARY

Robotics engineer with hands-on ownership of full-stack robotics systems — from architecture and simulation through embedded firmware, hardware integration, and deployment. Proficient in ROS/ROS2, C++, Python, and Linux, with strong experience in motion planning, kinematics, control systems, embedded systems, and robotic perception. Has independently driven system architecture decisions, led cross-functional coordination with mechanical, electronics, and software teams, and taken robotic systems from concept and prototype through working deployment. Comfortable owning technical direction and mentoring engineers on debugging, design review, and system-level integration.

CORE TECHNICAL SKILLS

Robotics Software: ROS 2 (Humble/Jazzy), Nav2, RTAB-Map, MoveIt 2, TF/URDF, SLAM, localization, motion planning, forward/inverse kinematics, trajectory planning, control systems

Languages & Platforms: C++, Python, Embedded C, Linux, Git

Simulation / Digital Twin: NVIDIA Isaac Sim, Isaac Lab, PyBullet, Gazebo, reinforcement learning (PPO)

Embedded Deployment: Teensy 4.1, STM32, ESP32/ESP8266, Raspberry Pi, PWM, PID, FOC concepts

Perception & Vision: LiDAR, RGB-D cameras, IMU, encoders, OpenCV, YOLOv8, MediaPipe, camera-to-robot calibration, sensor fusion

Hardware & Communication: Motor/servo drivers, UART, I2C, SPI, CAN, RS-485/Modbus, BLE, MQTT/HTTP, PCB and control-panel integration

Tools: SolidWorks, STM32CubeIDE/CubeMX, Keil, Proteus, easyEDA, ST-Link

PROFESSIONAL EXPERIENCE

Robotics Engineer — WeVois Labs Pvt. Ltd.

Feb 2026 - Present

Sole technical owner of a high-speed, vision-guided Delta robot pick-and-place system — architecture, firmware, electronics, and integration, from scratch through final deployment.

Robotics Engineer — HighFlow Industries Pvt. Ltd. (NextStep Robotics)

Dec 2024 - Feb 2026

Owned the autonomous mobile-robot software stack and developed mapping and navigation workflows in ROS 2; designed and validated the AMR environment, 3D mapping, and navigation stack in NVIDIA Isaac Sim. Real-hardware AMR testing with 3D LiDAR/cameras was not part of this work.

Embedded & Robotics R&D Consultant / Contract — XPRT Enforcement Pvt. Ltd.

Aug 2022 - Feb 2023

Integrated GPS/radar telematics for real-time vehicle tracking and authored technical architecture documentation for municipal IoT deployment.

Embedded Engineer — Experio Traffiko System Pvt. Ltd.

Nov 2020 - Oct 2021

Delivered end-to-end embedded product development — circuit design, PCB layout, and SMD assembly — for AI-driven traffic-enforcement hardware (“Speedo Soft”) using ANPR-based violation detection. Owned firmware for radar-based speed detection, dual-battery intelligent power management, and breath-analyzer modules; managed the full hardware lifecycle from schematic to validated PCB.

KEY ROBOTICS PROJECTS

- **Autonomous Vision-Guided Delta Robot** — End-to-End Ownership (WeVois Labs): Took a high-speed Delta robot for vision-guided pick-and-place from scratch to full deployment as sole technical owner. Built the complete Teensy 4.1 motion-control firmware (custom Delta inverse kinematics, synchronized multi-axis control, fault recovery, safety sequencing); designed the circuit and PCB and handled assembly and control-panel wiring; built a PC-side robot SDK for motion commands, calibration, and system monitoring/control; and integrated camera-based perception (object detection, camera-to-robot calibration,

conveyor tracking). Drove the design migration from a stepper-based prototype to a production-grade AC servo architecture. Owned complete system integration end-to-end, guided the mechanical team on design requirements, and directed the vision team's scope of work.

- **Scratch-Built Autonomous AMR** — Simulation & Navigation: Designed the robot in SolidWorks, built the URDF and simulation environment, and integrated ROS 2, RTAB-Map, Nav2, and 3D LiDAR for 3D point-cloud and occupancy-map generation. Validated the mapping and navigation stack in NVIDIA Isaac Sim; real-hardware testing with 3D LiDAR/cameras was not performed.
- **RL-Based Robotic Arm Training**: Ran massively-parallel PPO training in Isaac Sim/Isaac Lab; designed reward/observation logic and validated policies via ROS 2 bridge.
- **Autonomous Vision-Based Pick & Place**: Built full perception-to-actuation pipeline — YOLO/OpenCV detection, camera-to-robot calibration, inverse kinematics, trajectory planning.
- **Humanoid Robotic Hand & Gesture Control**: Multi-DOF finger/arm control via IMU/vision input, synchronized with a simulation digital twin.
- **QDD Actuator Design & Testing**: Designed and validated a quasi-direct-drive actuator (BLDC + planetary transmission); evaluated back-drivability, thermal performance, backlash.
- **Bipedal Robot Walking Simulation**: Implemented gait generation and stability-focused trajectory control in PyBullet using IK and PID/MPC concepts.

Additional applied projects: *intelligent power management systems, LiDAR data acquisition, embedded IoT/automation prototypes, and vehicle telemetry systems — details available on request.*

LEADERSHIP & DELIVERY

- Independently owns system architecture end-to-end on projects spanning simulation, electronics, firmware, software, hardware integration, and deployment.
- Regularly serves as the technical point of coordination across mechanical, electronics, vision, and software functions on multidisciplinary robotics programs.
- Reviews designs, root-causes cross-domain failures, and documents systems for repeatable testing and deployment.

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University | B.Tech, Electronics & Communication Engineering — 2021 - 2024

Board of Technical Education, Uttar Pradesh | Diploma, Electronics Engineering — 2018 - 2020

TRAINING

IoT Training, Softpro India (2024) | Python with IoT & Raspberry Pi, Softpro India (2020) | PCB Design, Softpro India (2020) | Embedded Systems & OS, ITI Limited Mankapur (2019)